

Floating Point

COMP211 Connor McMahon

This Week

- Information on next week's debugging checkoff can be found here.
  - Make sure to sign up for a slot by Friday!
  - Debugging Prep Sessions
    - Tomorrow (Thursday) 4-5pm
    - Tomorrow (Thursday) 5-6pm

- Homework 4 (Debugging) is due Friday.

- The TAs and I will be grading the quizzes tomorrow. We plan to get your grades back to you by the end of the week.

UNC CS CAREER PANEL

WHAT IS AN AI ENGINEER?

WED | SEP 23 | 6:00 PM |SN014

Find out what it takes to build AI as a career.

PIZZA!



SHERARD GRIFFIN
SENIOR DIR, A.I ENGINEERING, RED HAT



DANIEL PAQUETTE
ENGINEERING MANAGER, DIRECT SUPPLY



RIDA BAYRAKTAR
AI ENGINEER II, TANIUM
UNC CS ALUM



JAKE TERRILL
AI ENGINEER, IBM
UNC CS ALUM

HOSTED BY AI@UNC

Computer Science

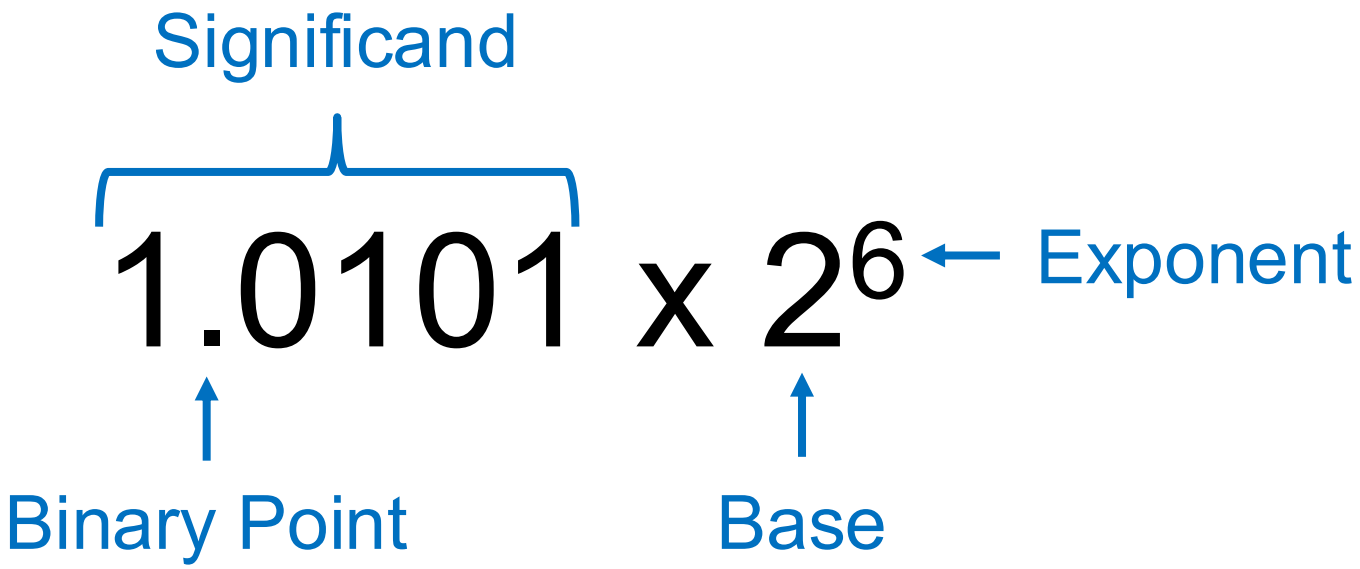
Floating Point Representation

- What is **floating point representation**?
  - A way to represent very large and very small numbers efficiently

- A standard called **IEEE-754** defines how modern systems handle these values.

- We will start with simpler models (**microfloat** and **minifloat**) to build up to **IEEE-754.**

Binary in Normalized Form



In normalized form, 1 <= |significand| < 2

This means the value to the left of the binary point is always 1.
We can increase the precision by not storing this 1!

Implicit 1

- In normalized binary floating point, every (nonzero) number has a leading 1 before the binary point.

- Since that leading one is always there, we don’t need to store it
  - Hence the name “Implicit 1”

- By not storing it, we free up one extra bit for the significand
  - This extra bit increase **precision**

- The **range** is unaffected by this
  - (Since the range is determined by the exponent bits)

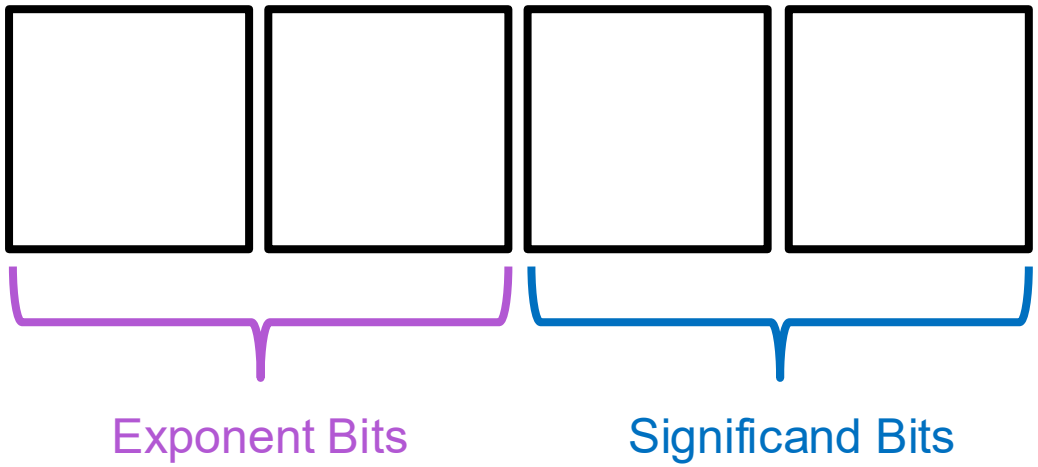
Microfloat V1

Floating Point

- The full value is not stored directly
- Instead, we store
  - **Significand** (without the leading one)
  - **Exponent**
- This design allows for a much larger **range** of values
  - But it in turn trades off **precision**

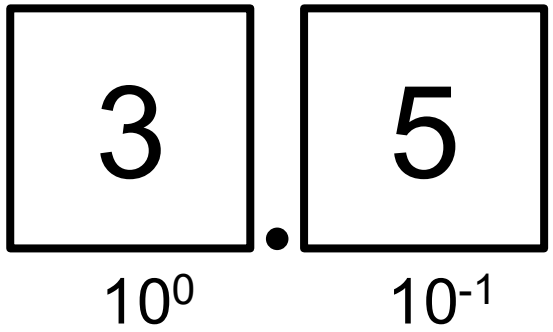
Microfloat

- We'll start by looking at a simple design called **microfloat**.
- Let's say we have 4 bits to represent our fractional value. Let's use two bits for the exponent and two bits for the significand.



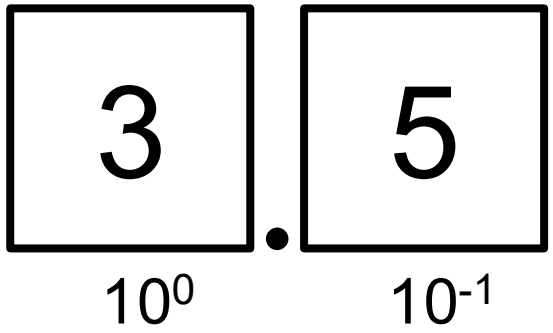
Microfloat V1: Encoding 3.5

Decimal Place-Value

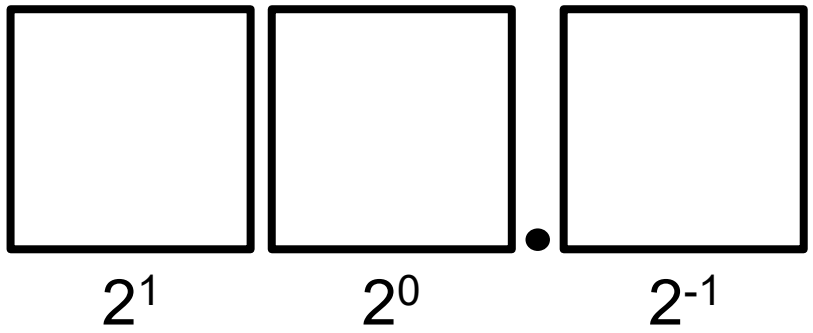


Microfloat V1: Encoding 3.5

Decimal Place-Value

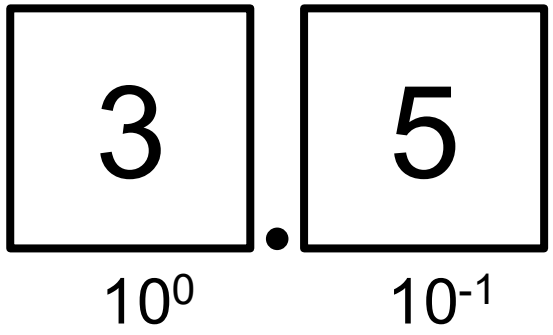


Binary Place-Value

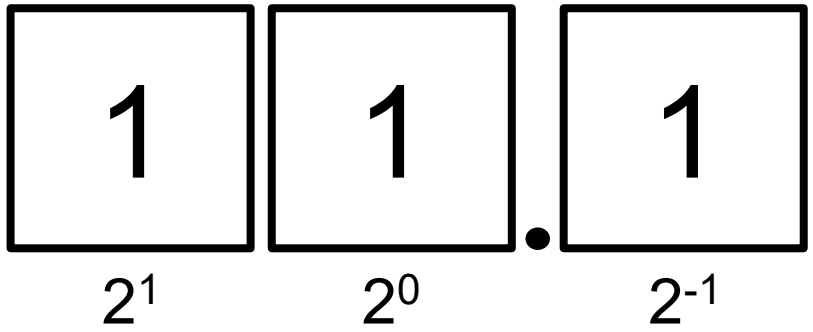


Microfloat V1: Encoding 3.5

Decimal Place-Value

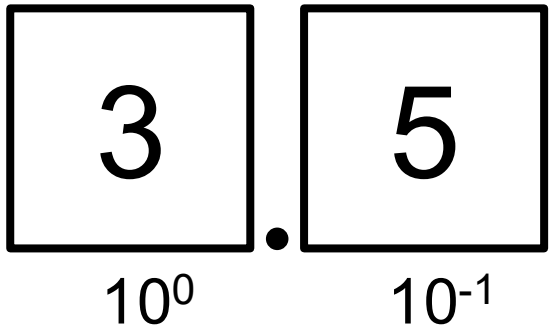


Binary Place-Value

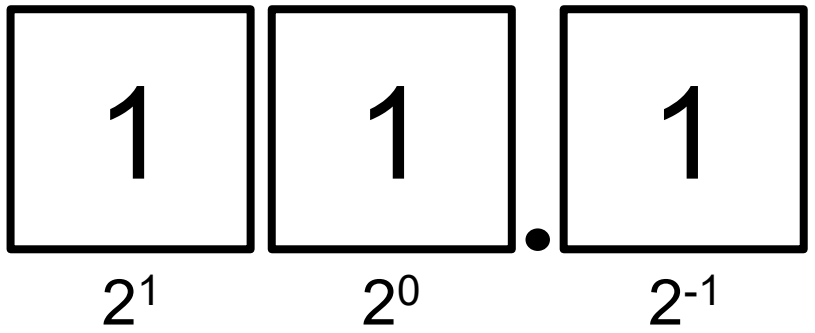


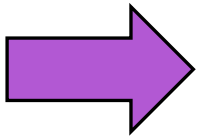
Microfloat V1: Encoding 3.5

Decimal Place-Value



Binary Place-Value





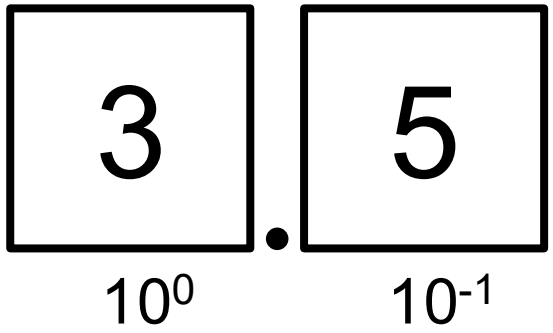
1.11×2^1

Exponent

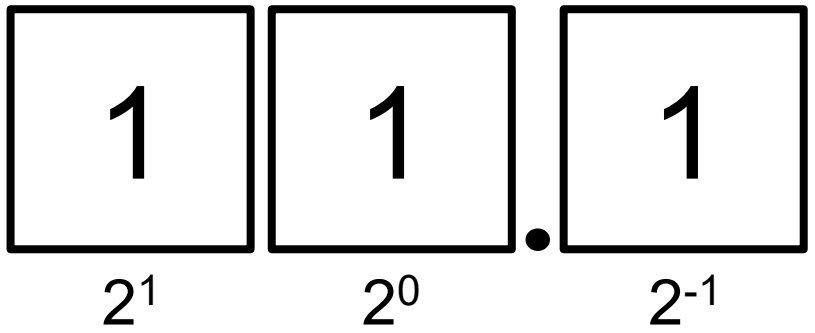
Significand
The leading 1 is not stored!

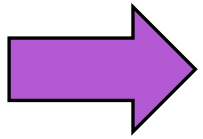
Microfloat V1: Encoding 3.5

Decimal Place-Value



Binary Place-Value



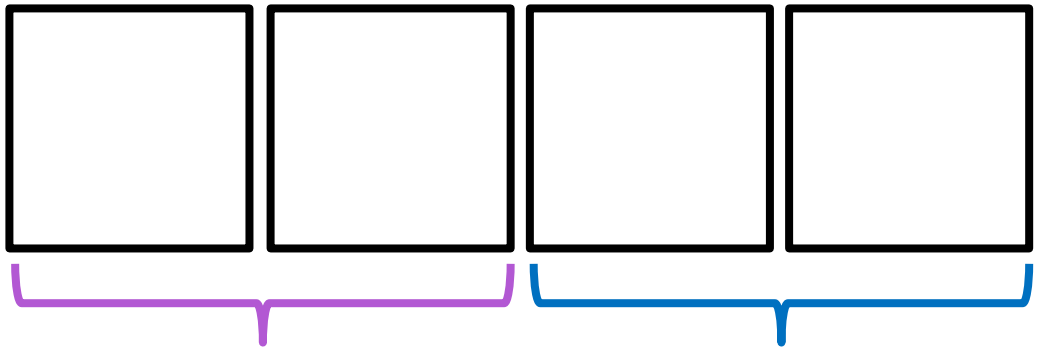


1.11×2^1

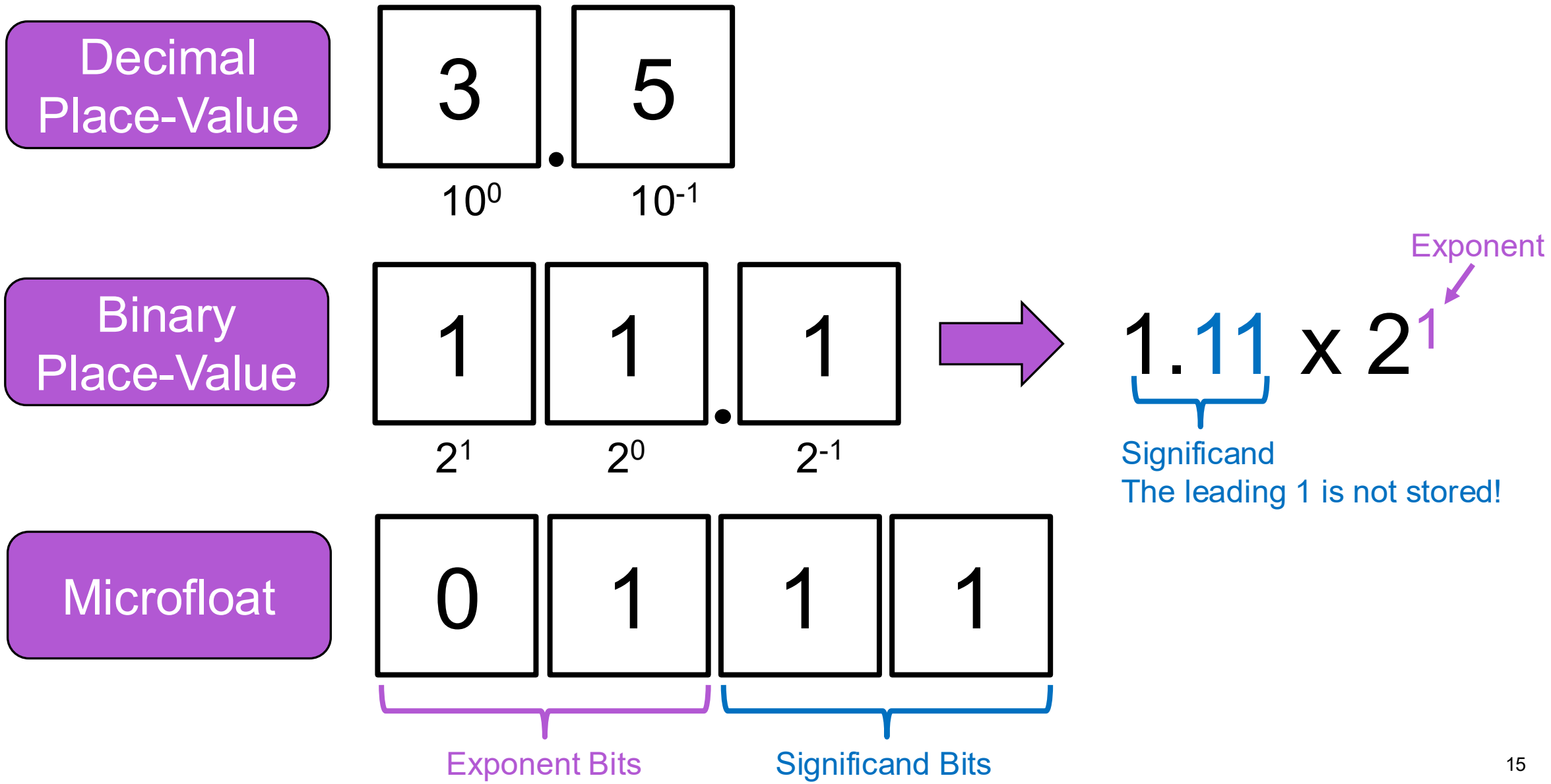
Exponent

Significand
The leading 1 is not stored!

Microfloat

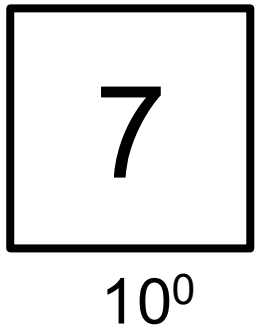


Microfloat V1: Encoding 3.5



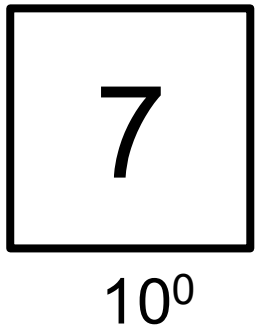
Microfloat V1: Encoding 7

Decimal Place-Value

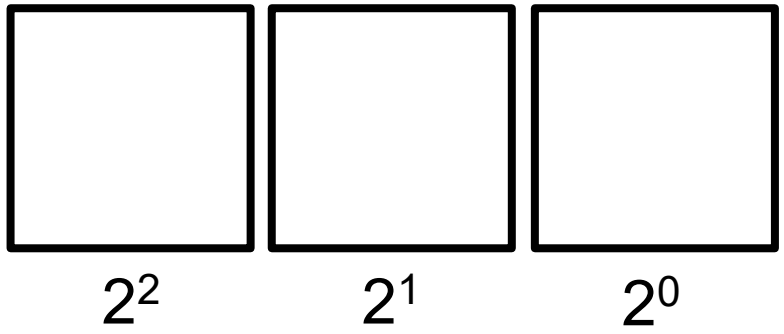


Microfloat V1: Encoding 7

Decimal Place-Value

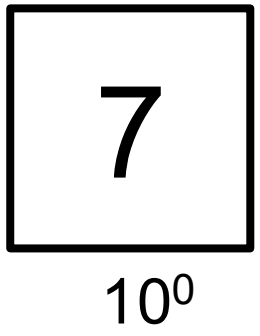


Binary Place-Value

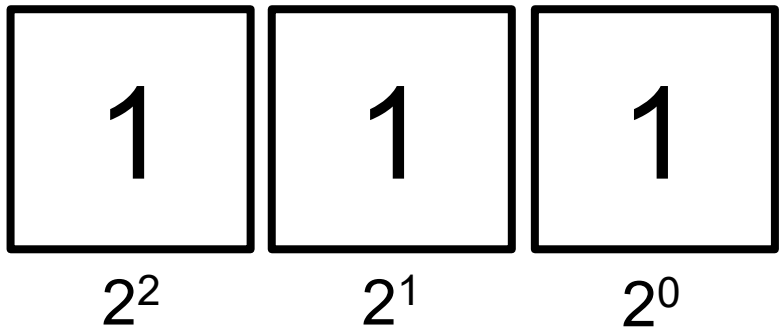


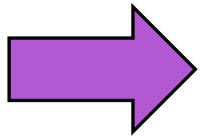
Microfloat V1: Encoding 7

Decimal Place-Value



Binary Place-Value





Exponent

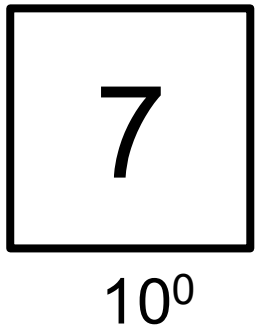
1.11×2^2

Significand

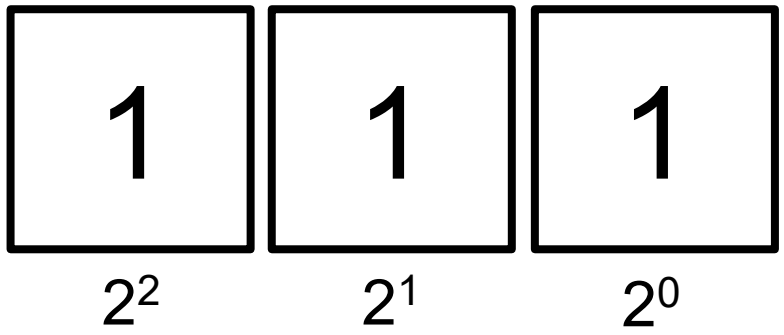
The leading 1 is not stored!

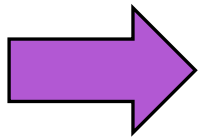
Microfloat V1: Encoding 7

Decimal Place-Value



Binary Place-Value



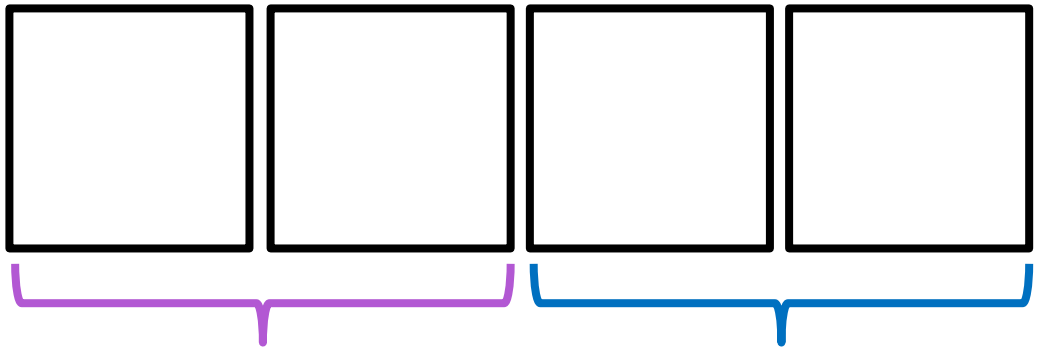


Exponent

1.11×2^2

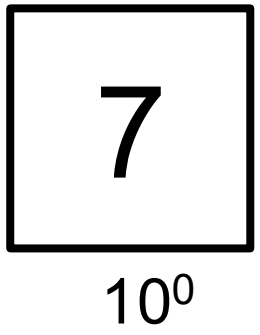
Significand
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Microfloat

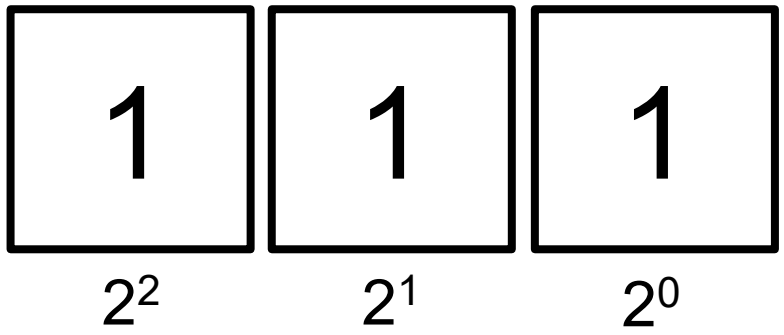


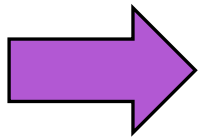
Microfloat V1: Encoding 7

Decimal Place-Value



Binary Place-Value



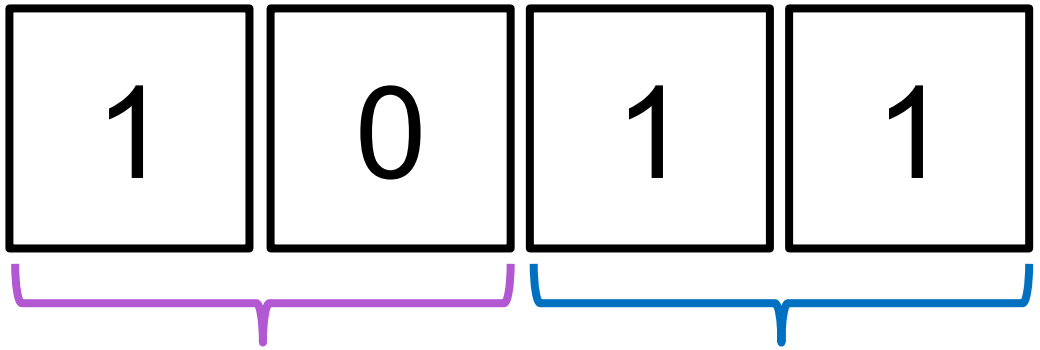


Exponent

1.11×2^2

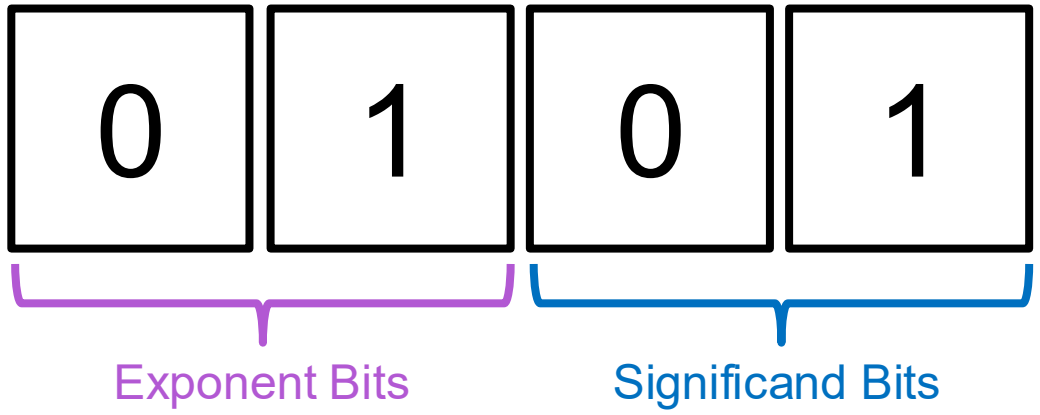
Significand
The leading 1 is not stored!

Microfloat

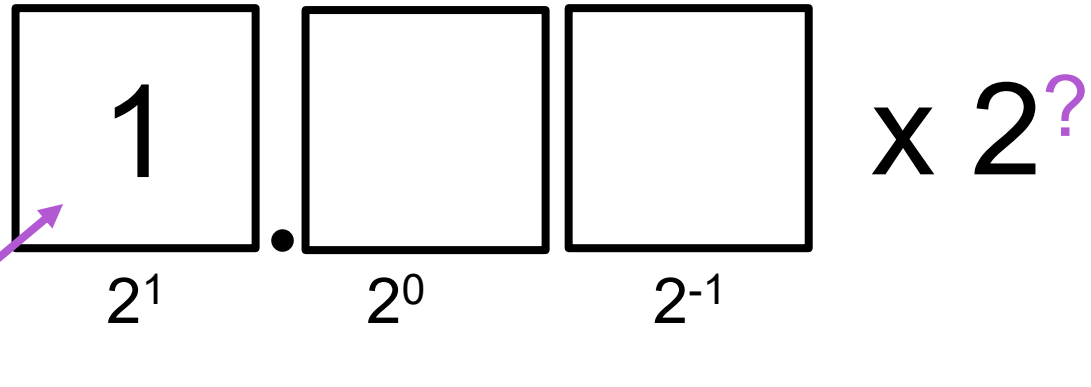


Microfloat to Decimal

Microfloat



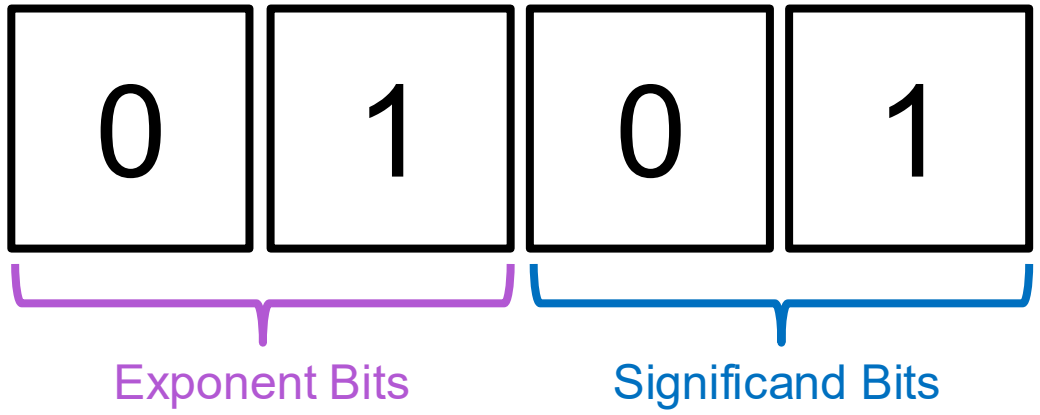
Binary



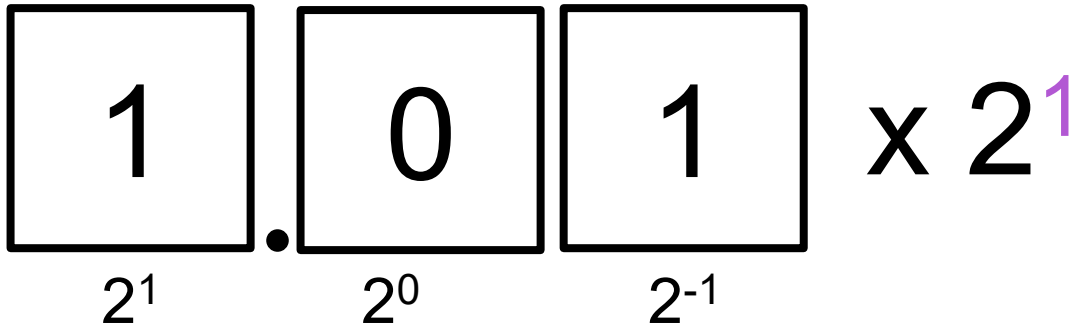
Implicit 1!

Microfloat to Decimal

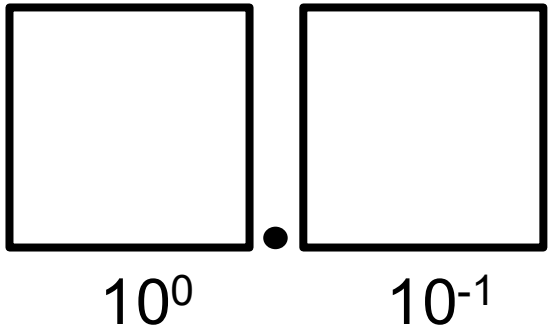
Microfloat



Binary

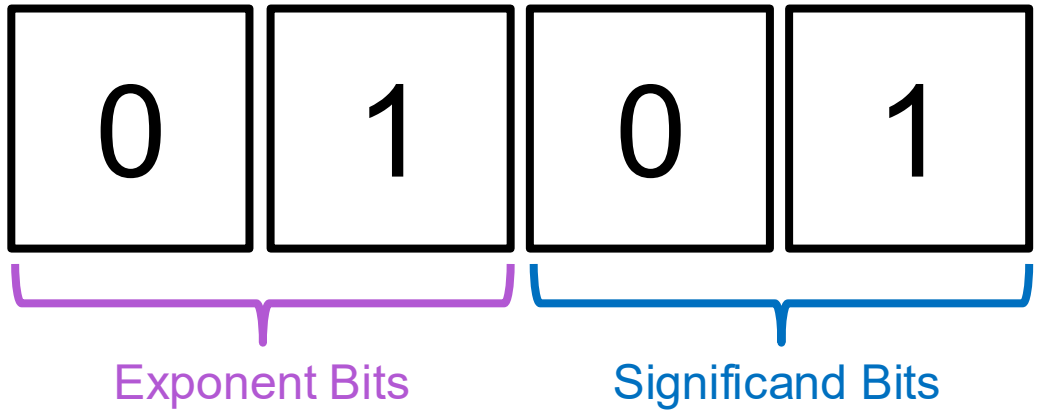


Decimal Place-Value

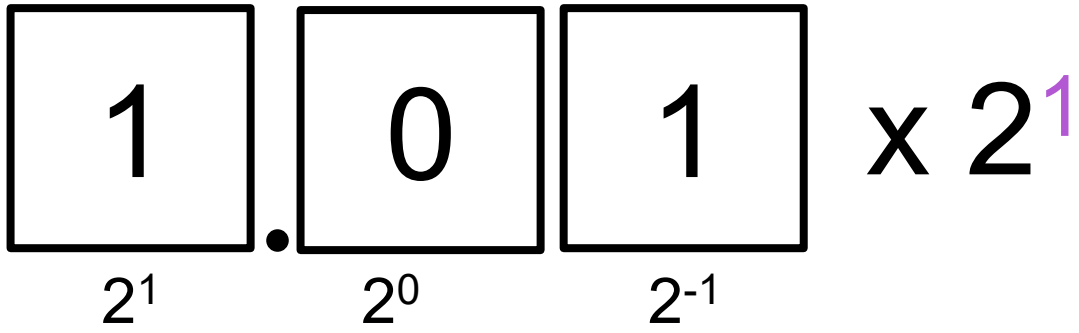


Microfloat to Decimal

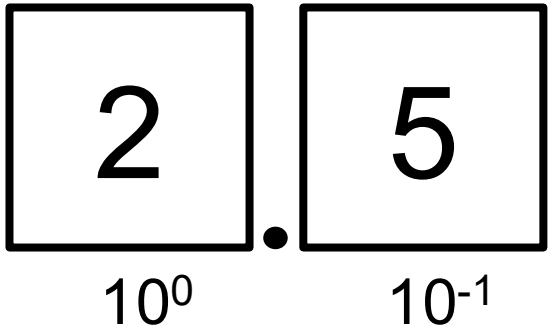
Microfloat



Binary



Decimal Place-Value



Microfloat V1

Exponent	Significand	Value	Decimal
00_2	00_2	$1.00_2 \times 2^0$	1
00_2	01_2	$1.01_2 \times 2^0$	1.25
00_2	10_2	$1.10_2 \times 2^0$	1.5
00_2	11_2	$1.11_2 \times 2^0$	1.75
01_2	00_2	$1.00_2 \times 2^1$	2
01_2	01_2	$1.01_2 \times 2^1$	2.5
01_2	10_2	$1.10_2 \times 2^1$	3
01_2	11_2	$1.11_2 \times 2^1$	3.5
10_2	00_2	$1.00_2 \times 2^2$	4
10_2	01_2	$1.01_2 \times 2^2$	5
10_2	10_2	$1.10_2 \times 2^2$	6
10_2	11_2	$1.11_2 \times 2^2$	7
11_2	00_2	$1.00_2 \times 2^3$	8
11_2	01_2	$1.01_2 \times 2^3$	10
11_2	10_2	$1.10_2 \times 2^3$	12
11_2	11_2	$1.11_2 \times 2^3$	14

Distance between consecutive values

0.25

0.5

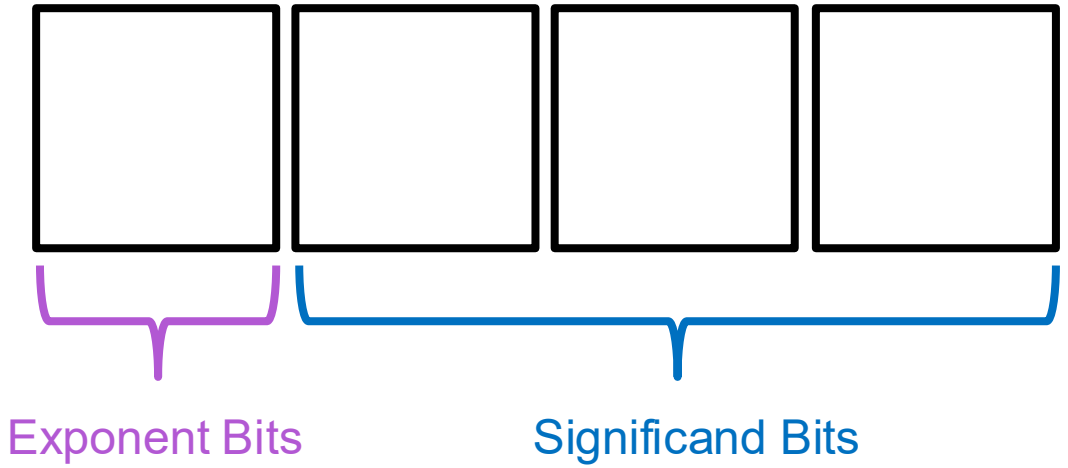
1

2

Microfloat V2

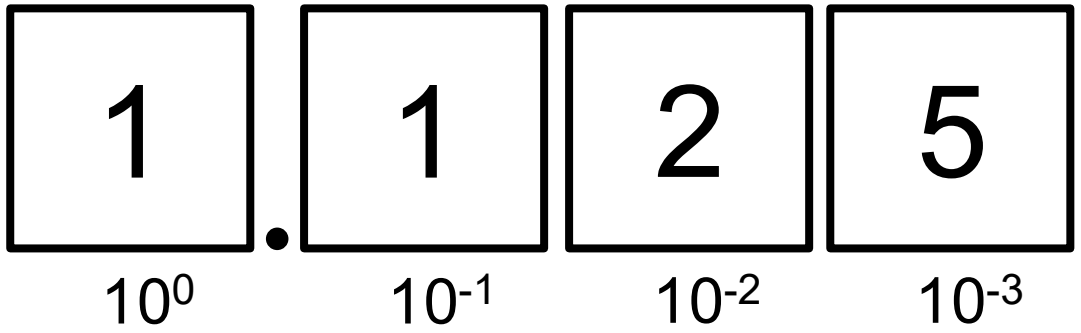
Microfloat V2

- Let's see what happens to the range and precision when we adjust the number of bits for the exponent and significand!



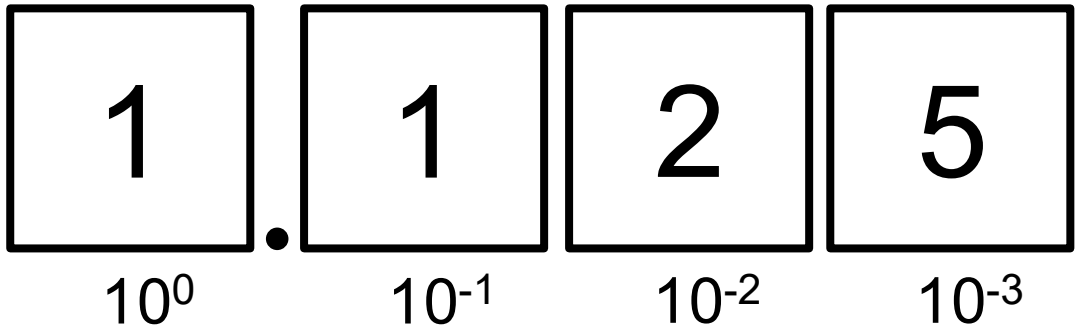
Microfloat V2: Encoding 1.125

Decimal Place-Value

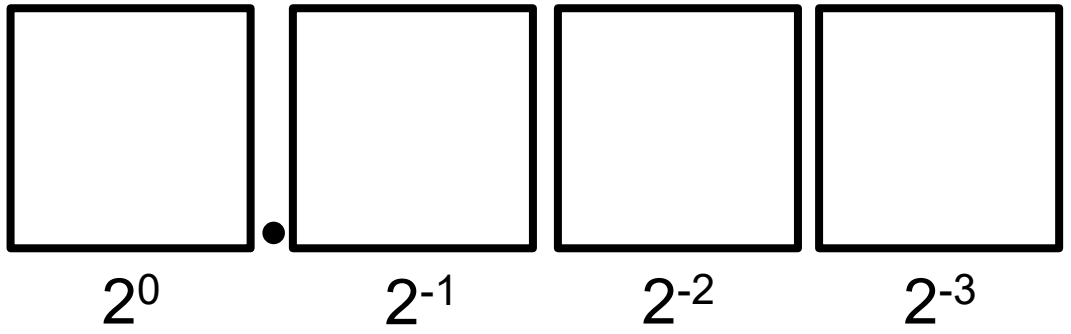


Microfloat V2: Encoding 1.125

Decimal Place-Value

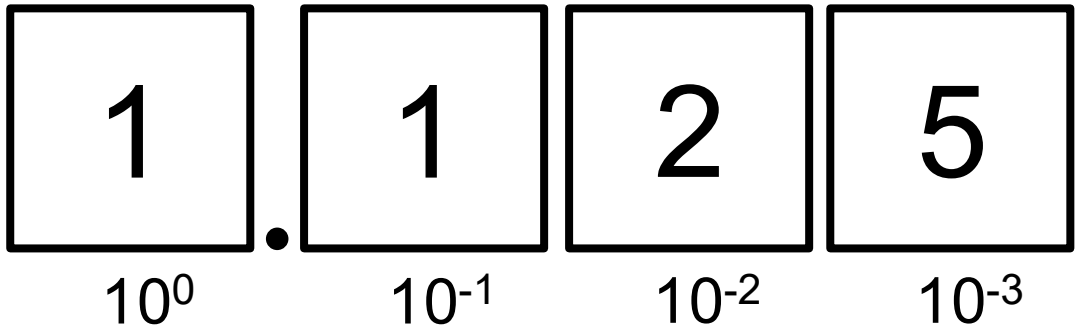


Binary Place-Value

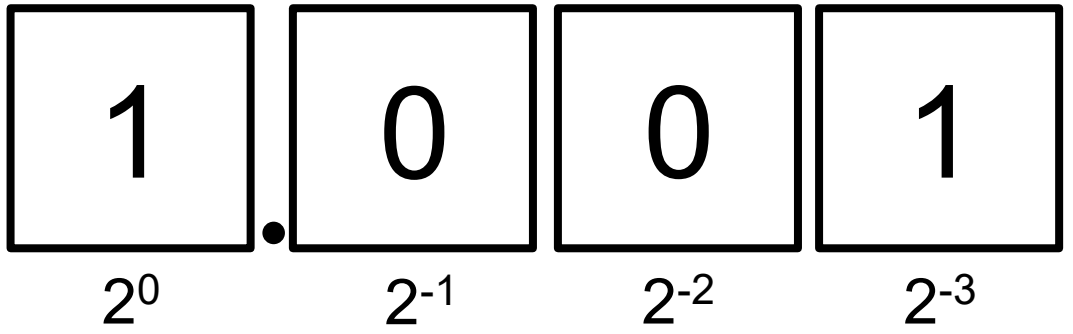


Microfloat V2: Encoding 1.125

Decimal Place-Value



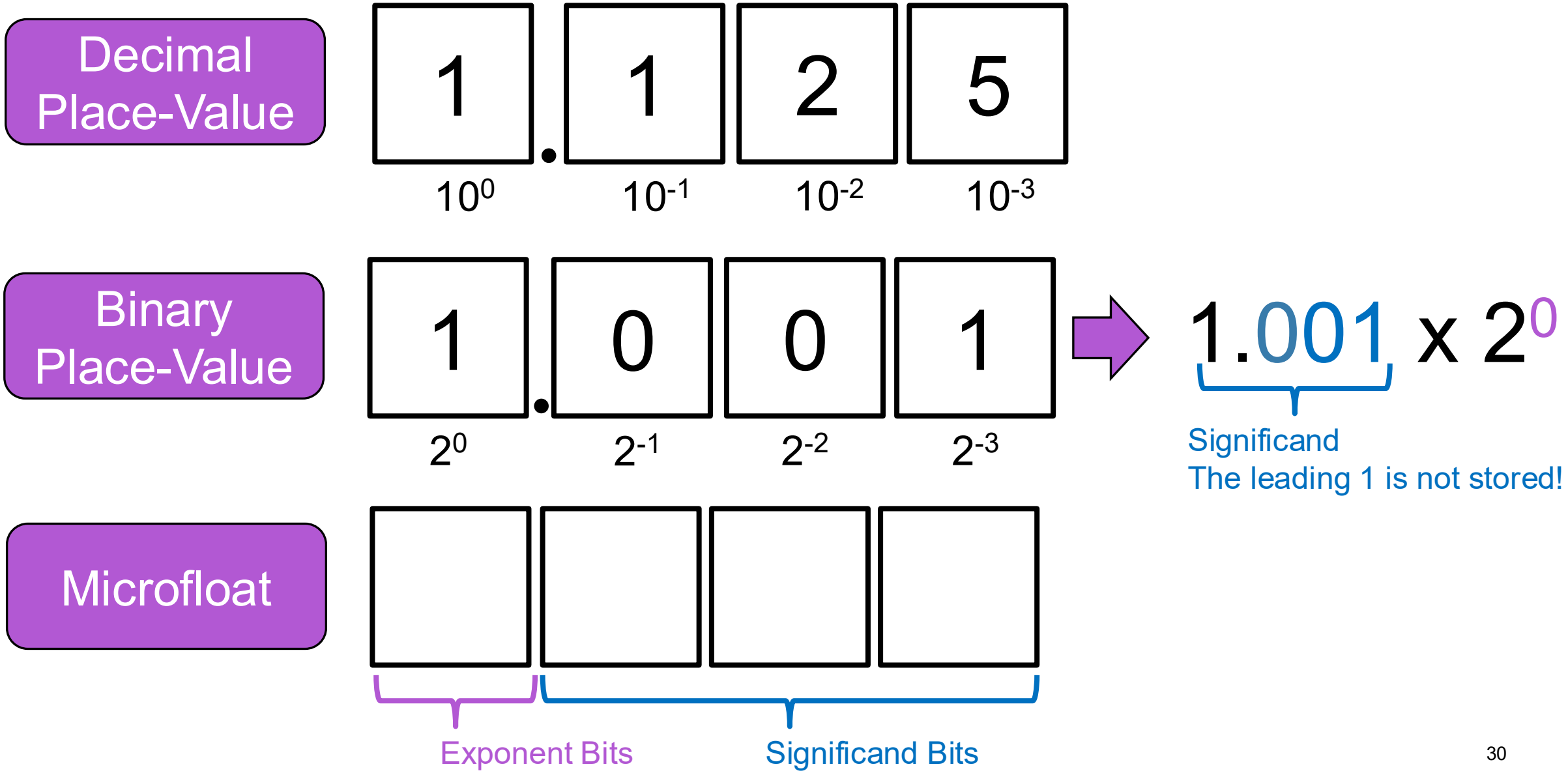
Binary Place-Value



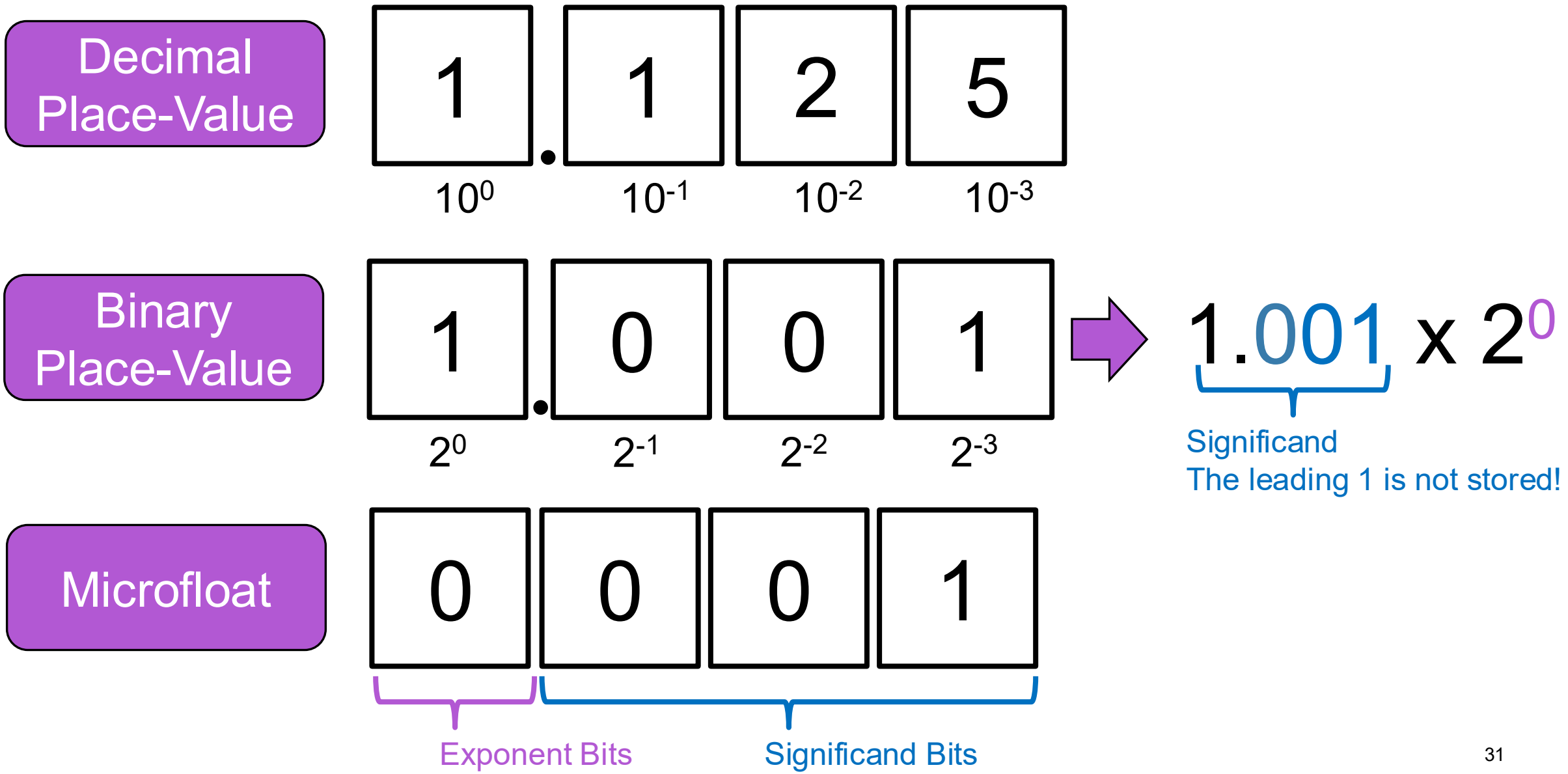
1.001×2^0

Significand
The leading 1 is not stored!

Microfloat V2: Encoding 1.125

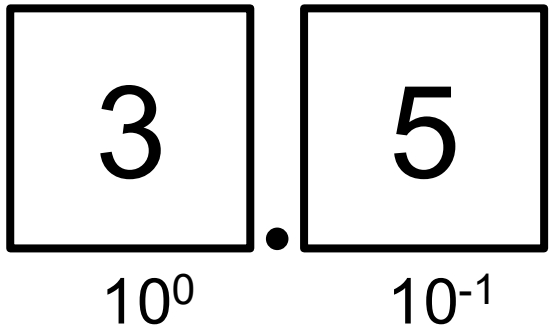


Microfloat V2: Encoding 1.125

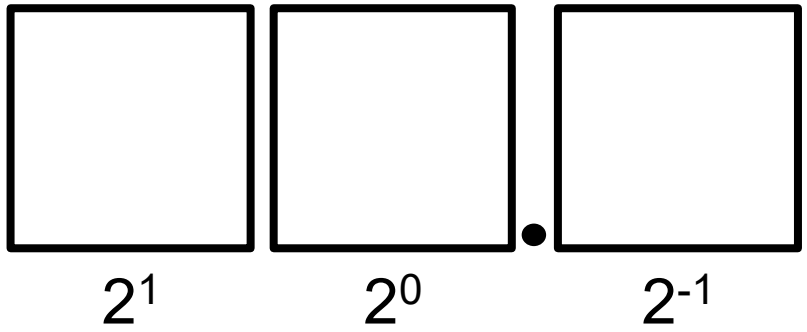


Microfloat V2: Encoding 3.5

Decimal Place-Value

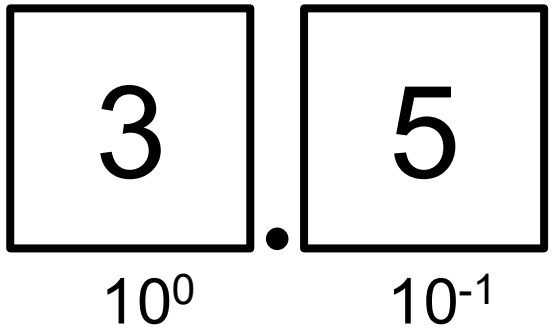


Binary Place-Value

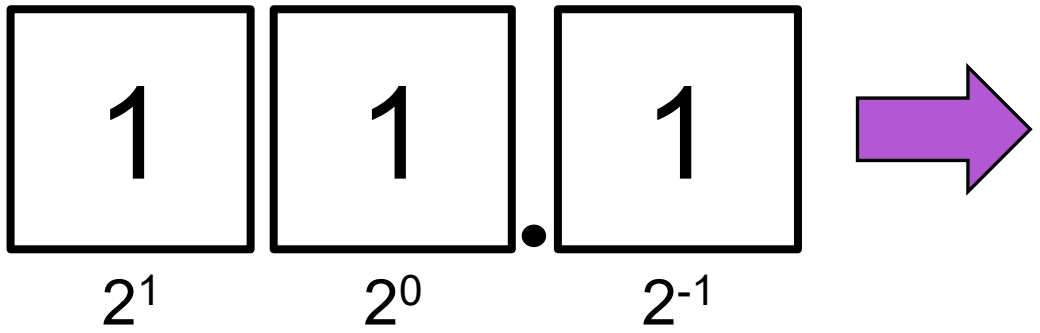


Microfloat V2: Encoding 3.5

Decimal Place-Value



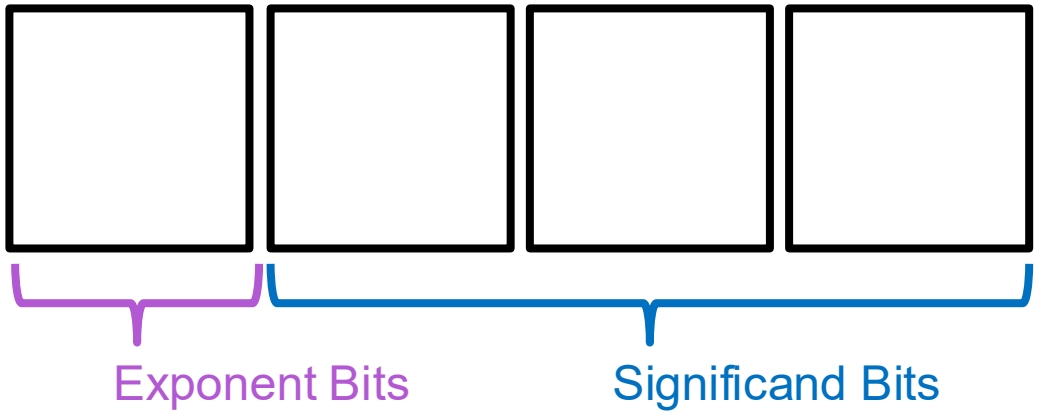
Binary Place-Value



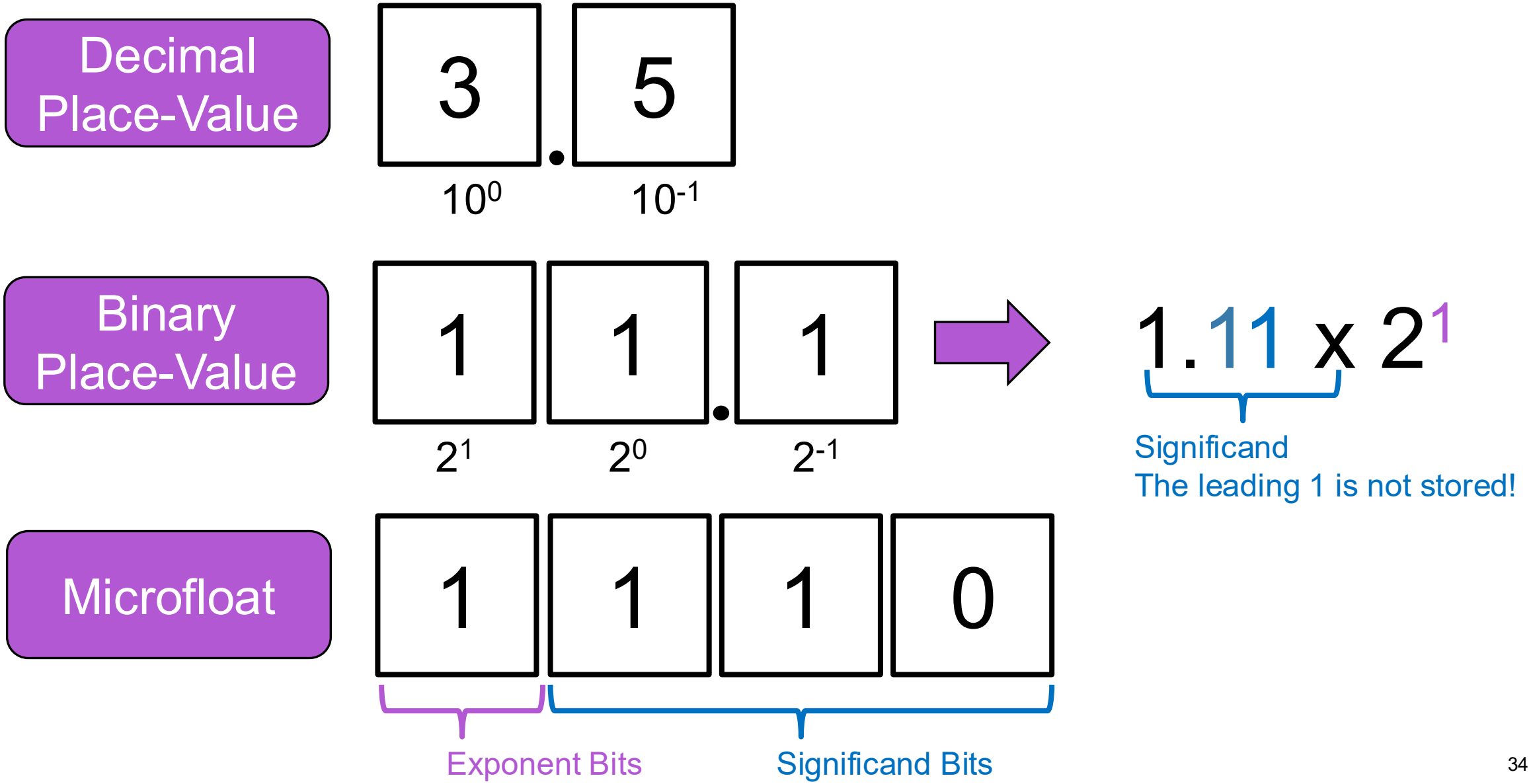
1.11×2^1

Significand
The leading 1 is not stored!

Microfloat



Microfloat V2: Encoding 3.5



Microfloat V2

Exponent	Significand	Value	Decimal
0_2	000_2	$1.000_2 \times 2^0$	1
0_2	001_2	$1.001_2 \times 2^0$	1.125
0_2	010_2	$1.010_2 \times 2^0$	1.25
0_2	011_2	$1.011_2 \times 2^0$	1.375
0_2	100_2	$1.100_2 \times 2^0$	1.5
0_2	101_2	$1.101_2 \times 2^0$	1.625
0_2	110_2	$1.110_2 \times 2^0$	1.75
0_2	111_2	$1.111_2 \times 2^0$	1.875
1_2	000_2	$1.000_2 \times 2^1$	2
1_2	001_2	$1.001_2 \times 2^1$	2.25
1_2	010_2	$1.010_2 \times 2^1$	2.5
1_2	011_2	$1.011_2 \times 2^1$	2.75
1_2	100_2	$1.100_2 \times 2^1$	3.5
1_2	101_2	$1.101_2 \times 2^1$	3.25
1_2	110_2	$1.110_2 \times 2^1$	3.5
1_2	111_2	$1.111_2 \times 2^1$	3.75

Distance between consecutive values

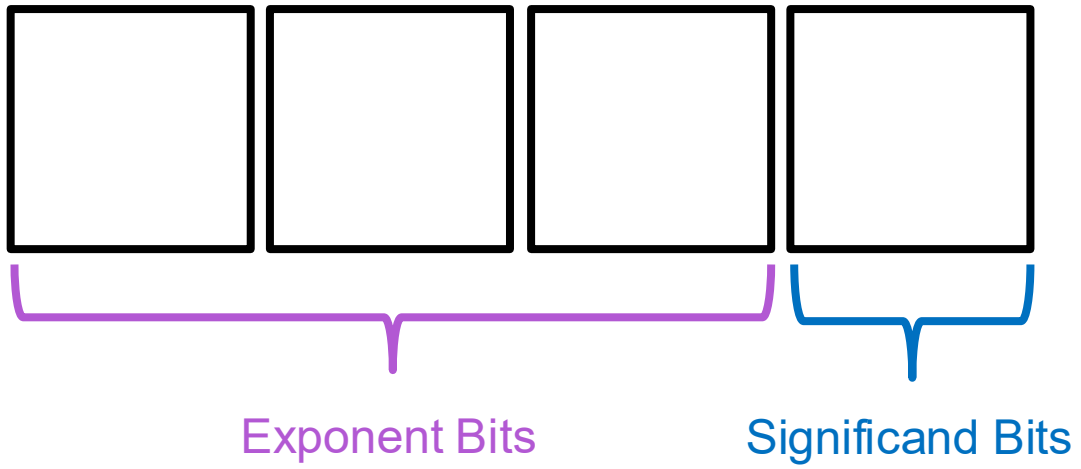
0.125

0.25

Microfloat V3

Microfloat V3

- Let's see what happens to the range and precision when we adjust the number of bits for the exponent and significand!

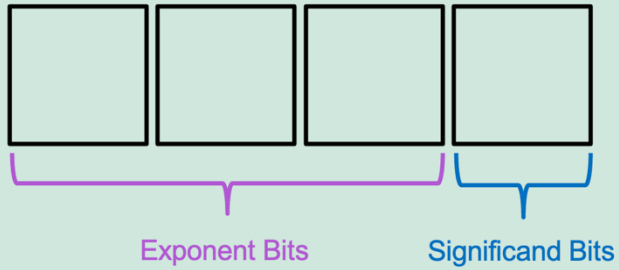


Microfloat V3

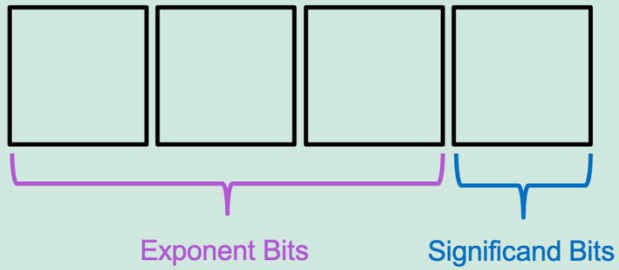
- For each value below, determine if it can be represented.
  - If it can be represented, write out its Microfloat V3 encoding.
  - If not, explain why: not enough precision or not enough range.
- A) 1.75 $1.11_2 \times 2^0$

  => not enough precision
- B) 3
- C) 192
- D) 20
- E) 256

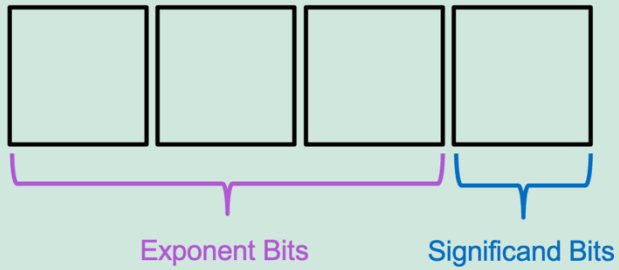
A



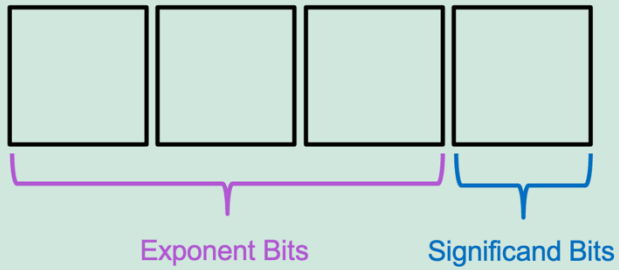
B



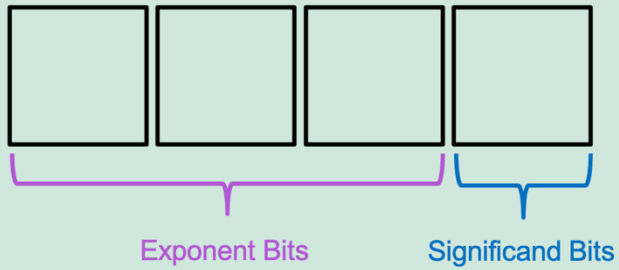
C



D



E

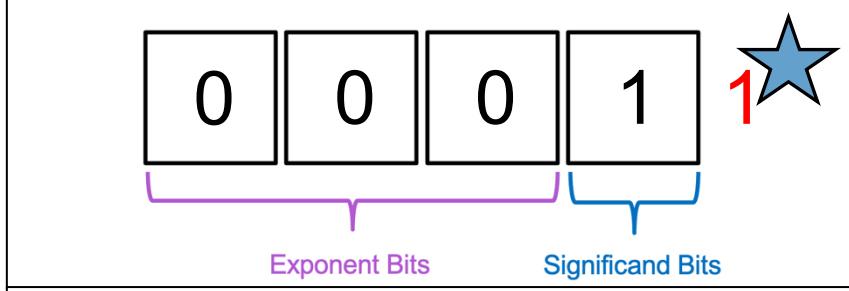


Microfloat V3

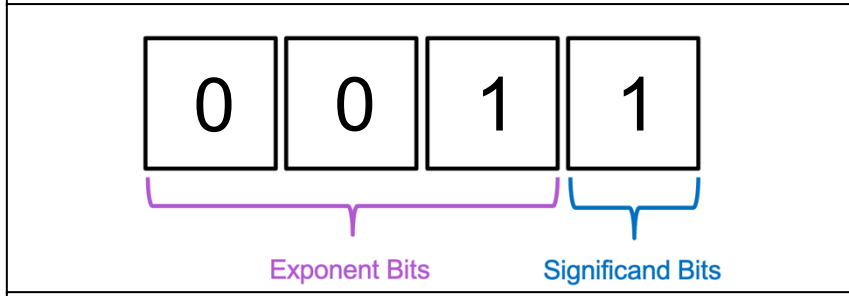
- For each value below, determine if it can be represented.
  - If it can be represented, write out its Microfloat V3 encoding.
  - If not, explain why: not enough precision or not enough range.
- A) $1.75 = 1.11_2 \times 2^0$

  => not enough precision
- B) $3 = 1.1_2 \times 2^1$ => 0011_2
- C) $192 = 1.1_2 \times 2^7$ => 1111_2
- D) $20 = 1.01_2 \times 2^4$ => not enough precision
- E) $256 = 1_2 \times 2^8$ => not enough range

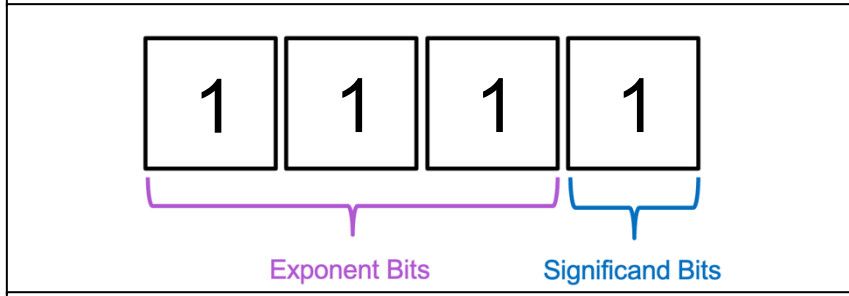
A



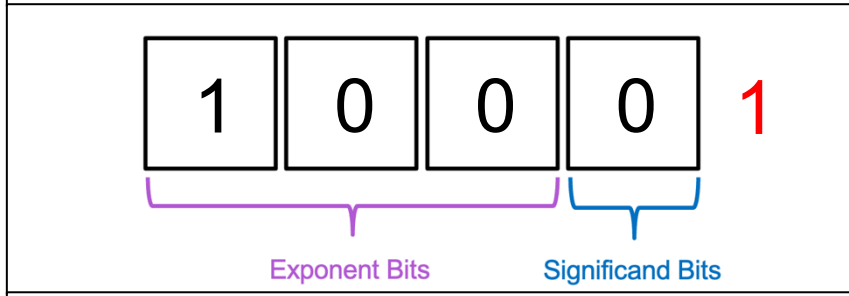
B



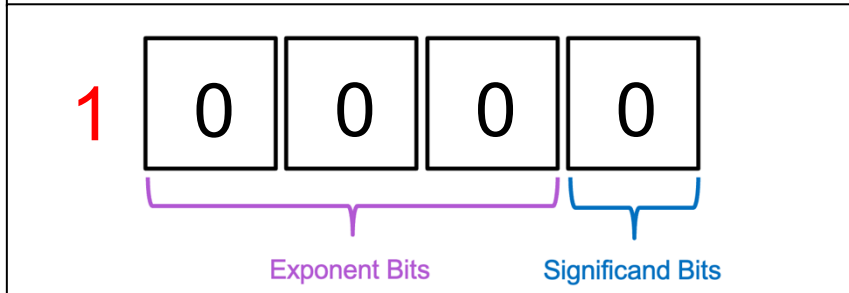
C



D



E



Microfloat V3

Exponent	Significand	Value	Decimal	Distance between consecutive values
000_2	0_2	$1.0_2 \times 2^0$	1	0.5
000_2	1_2	$1.1_2 \times 2^0$	1.5	
001_2	0_2	$1.0_2 \times 2^1$	2	1
001_2	1_2	$1.1_2 \times 2^1$	3	
010_2	0_2	$1.0_2 \times 2^2$	4	2
010_2	1_2	$1.1_2 \times 2^2$	6	
011_2	0_2	$1.0_2 \times 2^3$	8	4
011_2	1_2	$1.1_2 \times 2^3$	12	
100_2	0_2	$1.0_2 \times 2^4$	16	8
100_2	1_2	$1.1_2 \times 2^4$	24	
101_2	0_2	$1.0_2 \times 2^5$	32	16
101_2	1_2	$1.1_2 \times 2^5$	48	
110_2	0_2	$1.0_2 \times 2^6$	64	32
110_2	1_2	$1.1_2 \times 2^6$	96	
111_2	0_2	$1.0_2 \times 2^7$	128	64
111_2	1_2	$1.1_2 \times 2^7$	192	

Design Tradeoffs

- There's a tradeoff between how large or small a number you can represent (**range**) and how accurately you can represent numbers (**precision**), both of which are constrained by the total number of available bits.

- Precision vs Range
  - **More exponent bits increase the range** of representable numbers but reduces the number of bits available for the significand, **decreasing precision**.
  - **More significand bits increase the precision** of the representation, allowing for more precision, but reduce the number of bits available for the exponent, **decreasing the range**.

Microfloat Comparison

V1

Exponent	Significand	Value	Decimal	
00_2	00_2	$1.00_2 \times 2^0$	1	0.25
00_2	01_2	$1.01_2 \times 2^0$	1.25	
00_2	10_2	$1.10_2 \times 2^0$	1.5	
00_2	11_2	$1.11_2 \times 2^0$	1.75	
01_2	00_2	$1.00_2 \times 2^1$	2	0.5
01_2	01_2	$1.01_2 \times 2^1$	2.5	
01_2	10_2	$1.10_2 \times 2^1$	3	
01_2	11_2	$1.11_2 \times 2^1$	3.5	
10_2	00_2	$1.00_2 \times 2^2$	4	1
10_2	01_2	$1.01_2 \times 2^2$	5	
10_2	10_2	$1.10_2 \times 2^2$	6	
10_2	11_2	$1.11_2 \times 2^2$	7	
11_2	00_2	$1.00_2 \times 2^3$	8	2
11_2	01_2	$1.01_2 \times 2^3$	10	
11_2	10_2	$1.10_2 \times 2^3$	12	
11_2	11_2	$1.11_2 \times 2^3$	14	

V2

Exponent	Significand	Value	Decimal	
0_2	000_2	$1.000_2 \times 2^0$	1	0.125
0_2	001_2	$1.001_2 \times 2^0$	1.125	
0_2	010_2	$1.010_2 \times 2^0$	1.25	
0_2	011_2	$1.011_2 \times 2^0$	1.375	
0_2	100_2	$1.100_2 \times 2^0$	1.5	
0_2	101_2	$1.101_2 \times 2^0$	1.625	
0_2	110_2	$1.110_2 \times 2^0$	1.75	
0_2	111_2	$1.111_2 \times 2^0$	1.875	
1_2	000_2	$1.000_2 \times 2^1$	2	0.25
1_2	001_2	$1.001_2 \times 2^1$	2.25	
1_2	010_2	$1.010_2 \times 2^1$	2.5	
1_2	011_2	$1.011_2 \times 2^1$	2.75	
1_2	100_2	$1.100_2 \times 2^1$	3.5	
1_2	101_2	$1.101_2 \times 2^1$	3.25	
1_2	110_2	$1.110_2 \times 2^1$	3.5	
1_2	111_2	$1.111_2 \times 2^1$	3.75	

V3

Exponent	Significand	Value	Decimal	
000_2	0_2	$1.0_2 \times 2^0$	1	0.5
000_2	1_2	$1.1_2 \times 2^0$	1.5	
001_2	0_2	$1.0_2 \times 2^1$	2	1
001_2	1_2	$1.1_2 \times 2^1$	3	
010_2	0_2	$1.0_2 \times 2^2$	4	2
010_2	1_2	$1.1_2 \times 2^2$	6	
011_2	0_2	$1.0_2 \times 2^3$	8	4
011_2	1_2	$1.1_2 \times 2^3$	12	
100_2	0_2	$1.0_2 \times 2^4$	16	8
100_2	1_2	$1.1_2 \times 2^4$	24	
101_2	0_2	$1.0_2 \times 2^5$	32	16
101_2	1_2	$1.1_2 \times 2^5$	48	
110_2	0_2	$1.0_2 \times 2^6$	64	32
110_2	1_2	$1.1_2 \times 2^6$	96	
111_2	0_2	$1.0_2 \times 2^7$	128	64
111_2	1_2	$1.1_2 \times 2^7$	192	

Microfloat V4

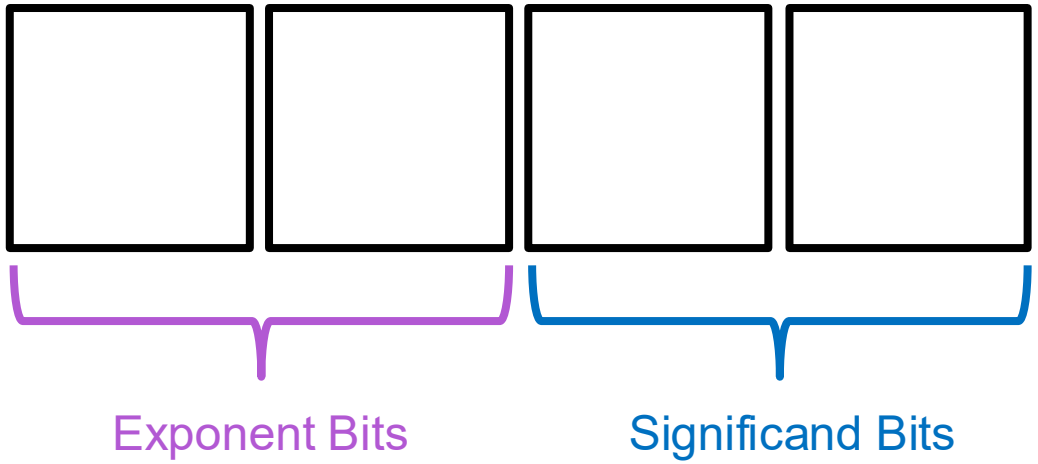
Representing Small Values

- **Problem:** Notice in the previous slide that we cannot represent values between 0 and 1
  - This is because the exponent must be positive

- **Solution:** Subtract a **fixed bias** from the exponent

Microfloat V4: Representing Small Values

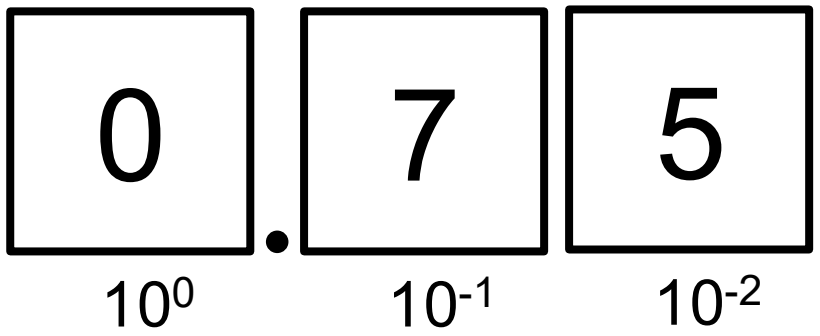
- Let's return back to having two exponent bits and two significand bits.



- Let's set the bias to 2. This makes the range of exponents [-2, 1]

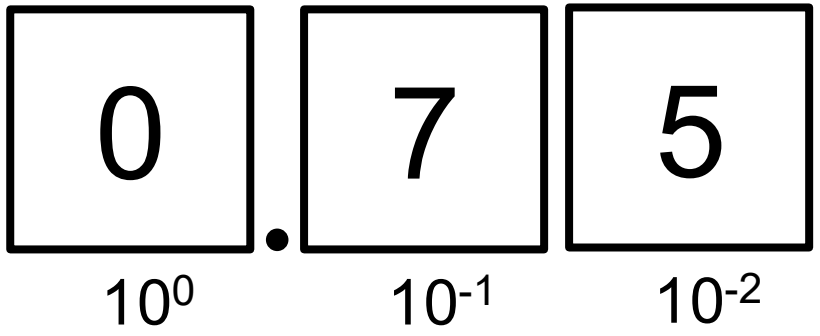
Microfloat V4: Encoding 0.75

Decimal Place-Value

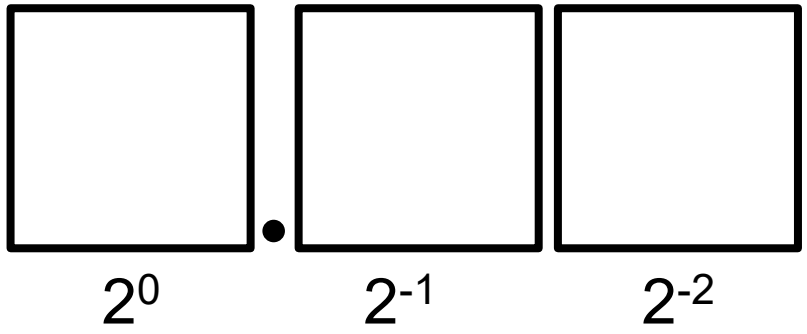


Microfloat V4: Encoding 0.75

Decimal Place-Value

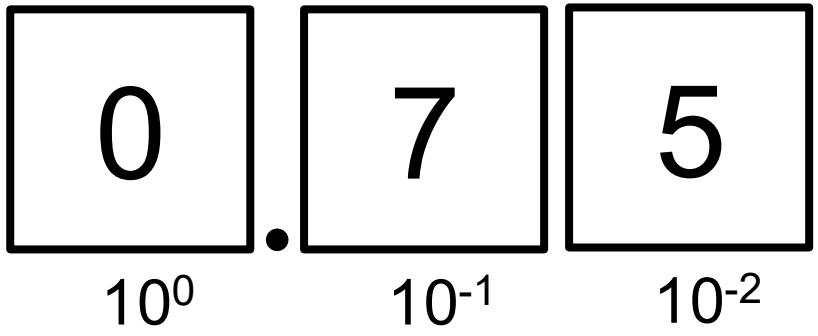


Binary Place-Value

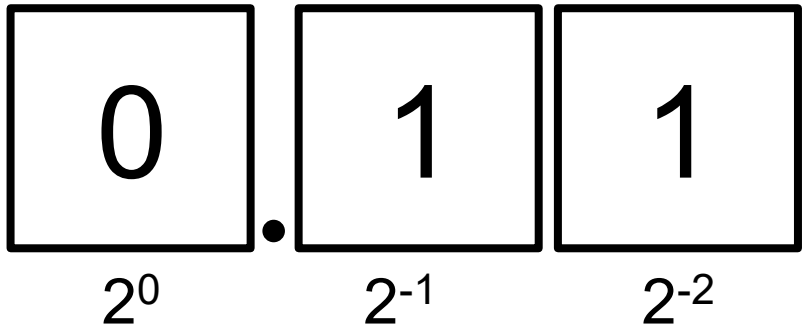


Microfloat V4: Encoding 0.75

Decimal Place-Value



Binary Place-Value

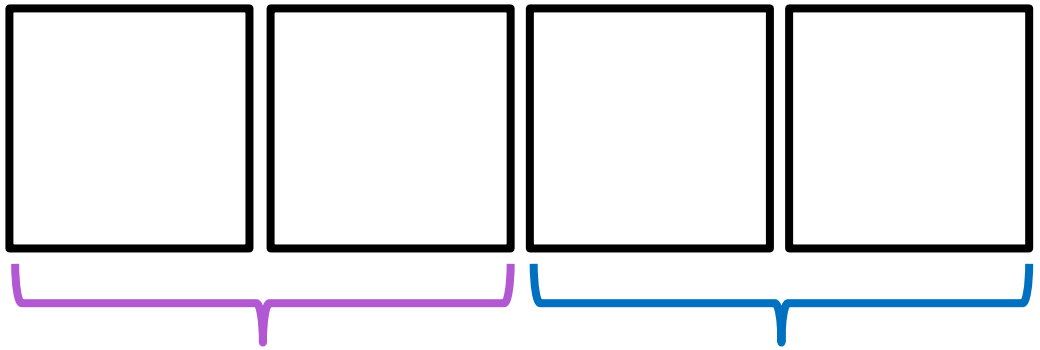


Exponent bits = -1 + bias = -1 + 2 = 1

1.1×2^{-1}

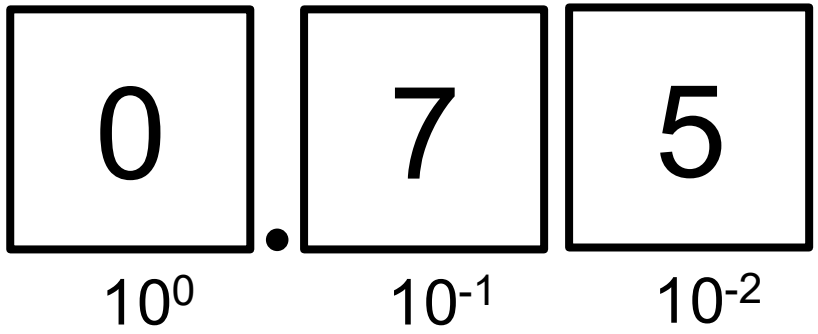
Significand
The leading 1 is not stored!

Microfloat

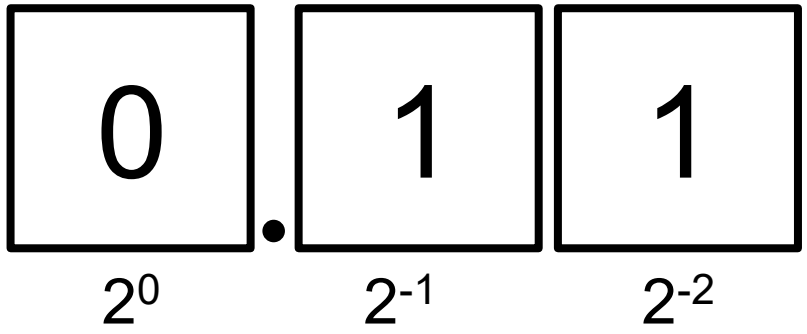


Microfloat V4: Encoding 0.75

Decimal Place-Value



Binary Place-Value

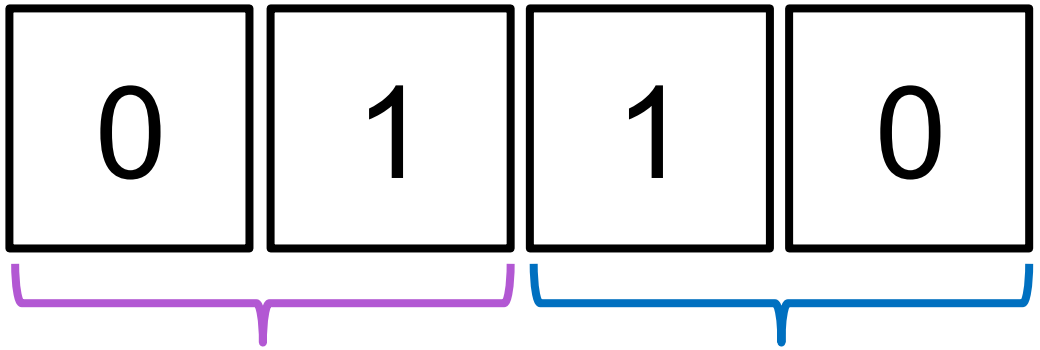


Exponent bits = -1 + bias = -1 + 2 = 1

1.1×2^{-1}

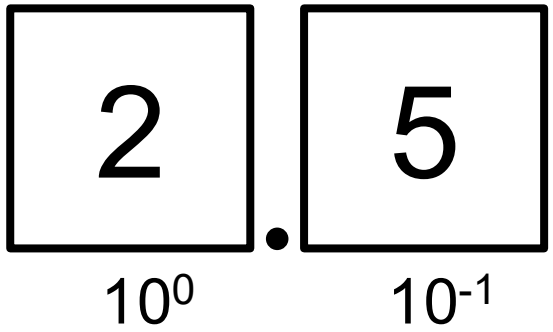
Significand
The leading 1 is not stored!

Microfloat



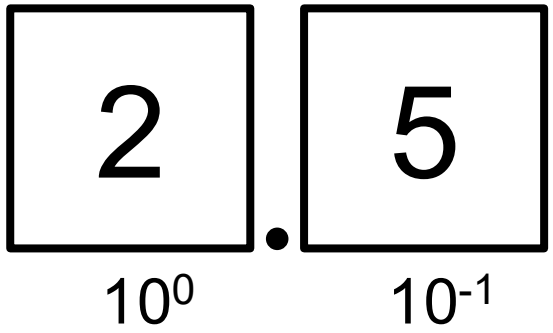
Microfloat V4: Encoding 2.5

Decimal Place-Value

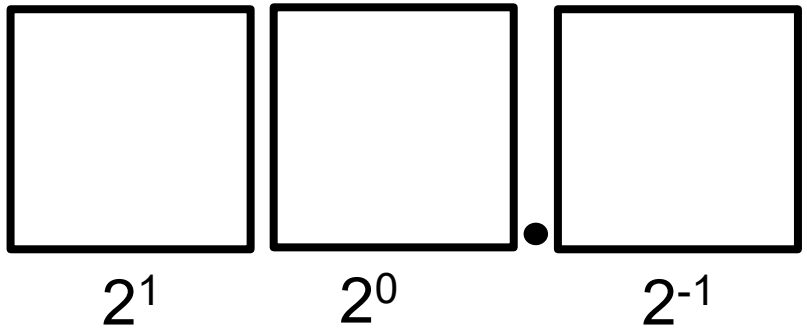


Microfloat V4: Encoding 2.5

Decimal Place-Value

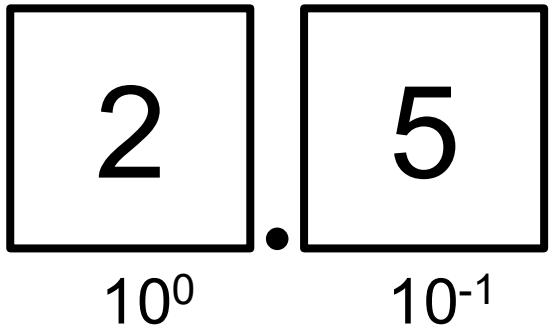


Binary Place-Value

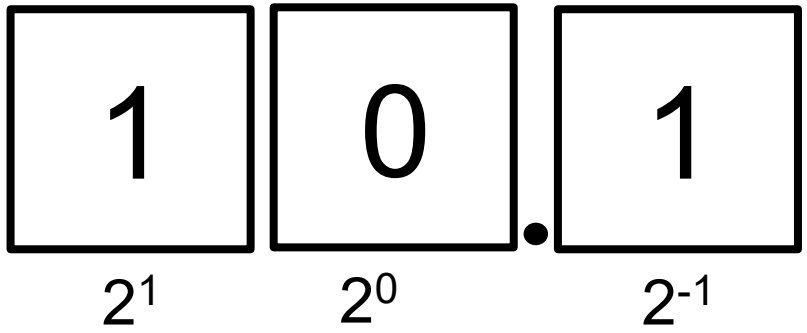


Microfloat V4: Encoding 2.5

Decimal Place-Value



Binary Place-Value

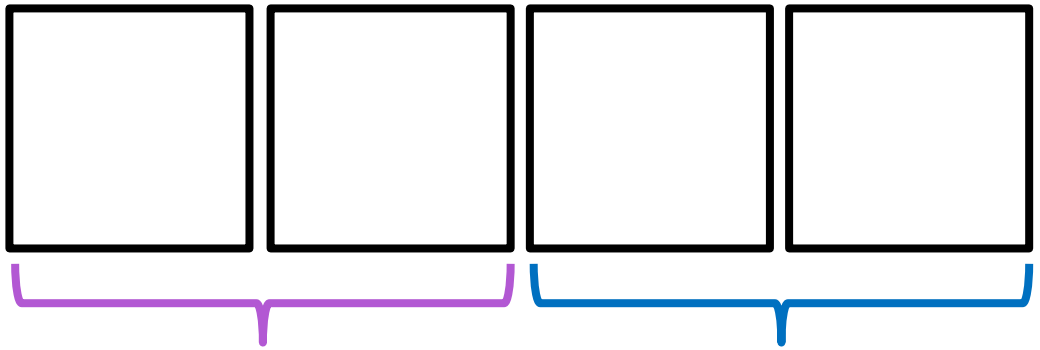


Exponent bits = 1 + bias = 1 + 2 = 3

1.01×2^1

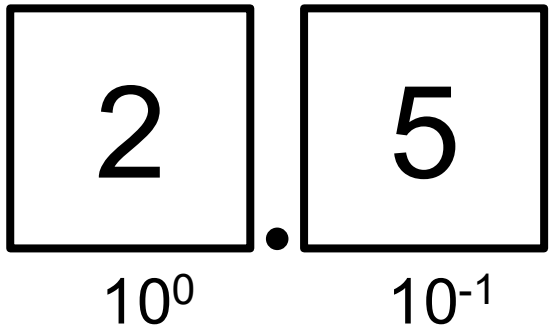
Significand
The leading 1 is not stored!

Microfloat

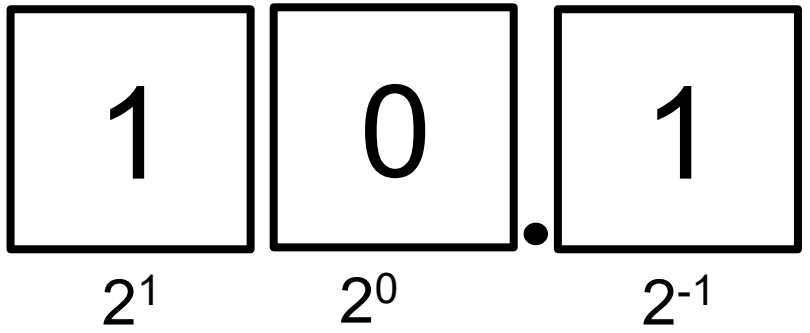


Microfloat V4: Encoding 2.5

Decimal Place-Value



Binary Place-Value

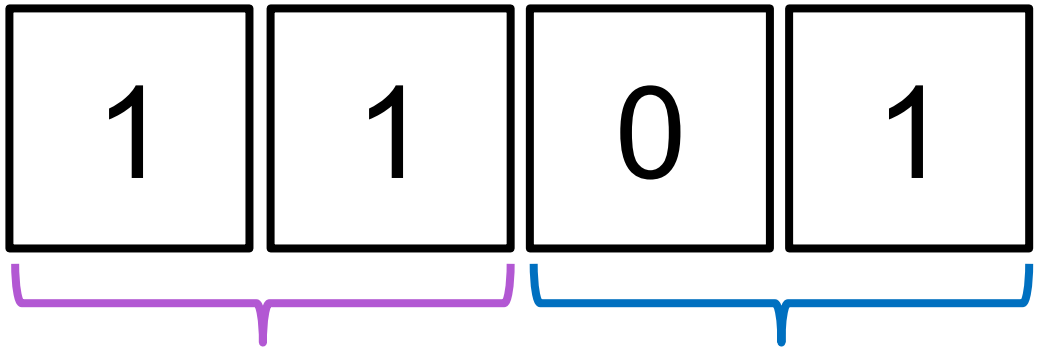


Exponent bits = 1 + bias = 1 + 2 = 3

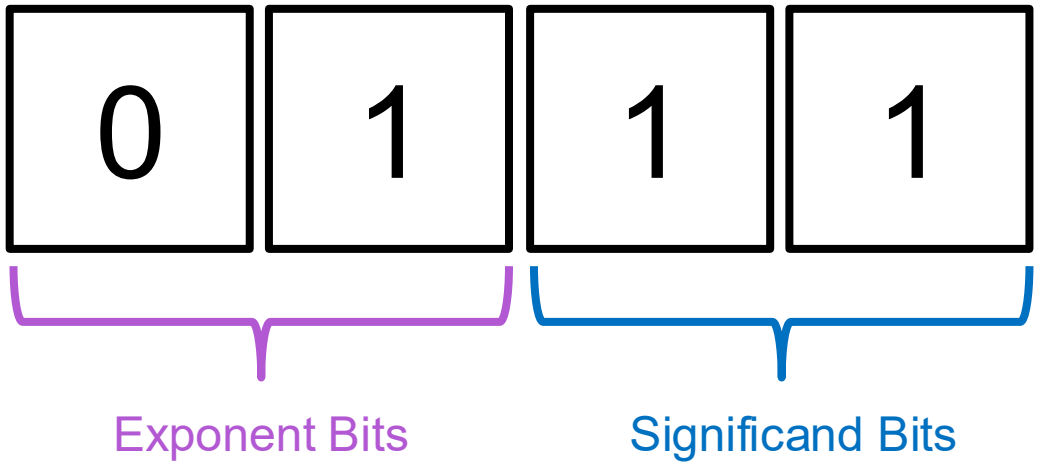
1.01×2^1

Significand
The leading 1 is not stored!

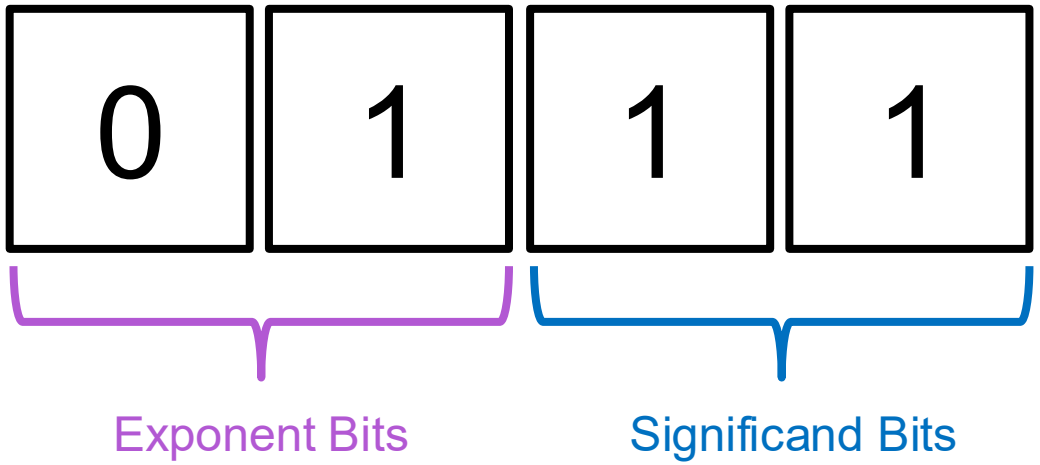
Microfloat



Example: Microfloat V4 to Decimal



Example: Microfloat V4 to Decimal



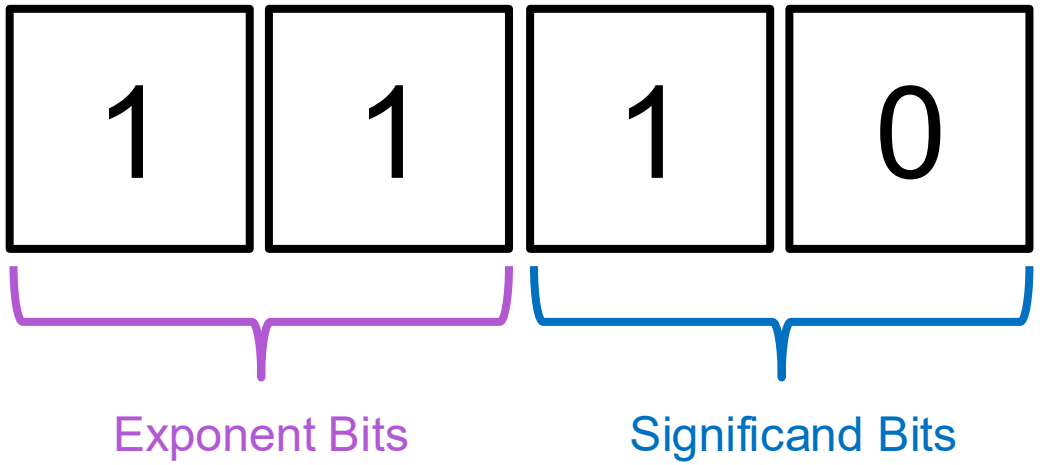
1.<significand bits> x $2^{<\text{exponent bits}>-\text{bias}}$

$1.11 \times 2^{1-2}$

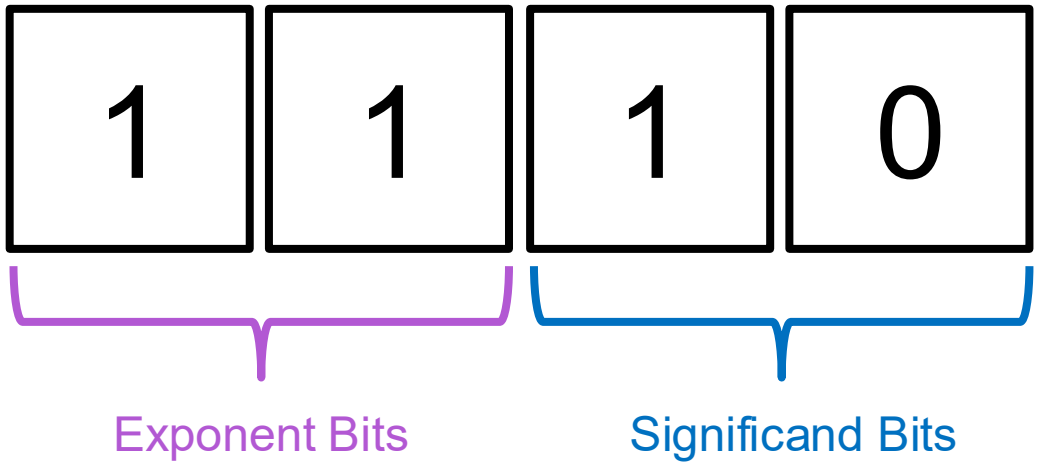
0.111

0.875

Example: Microfloat V4 to Decimal



Example: Microfloat V4 to Decimal



1.<significand bits> x $2^{<\text{exponent bits}>-\text{bias}}$

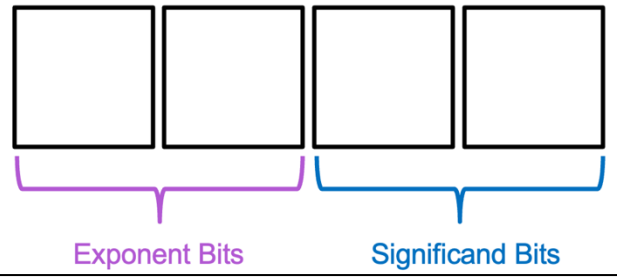
1.10x 2^{3-2}

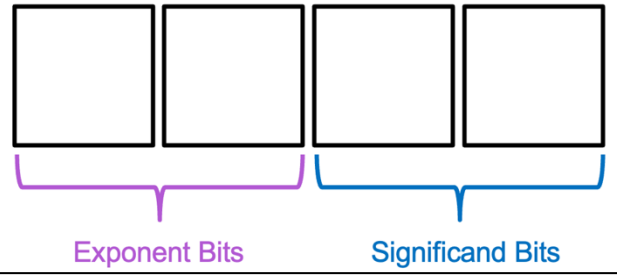
11

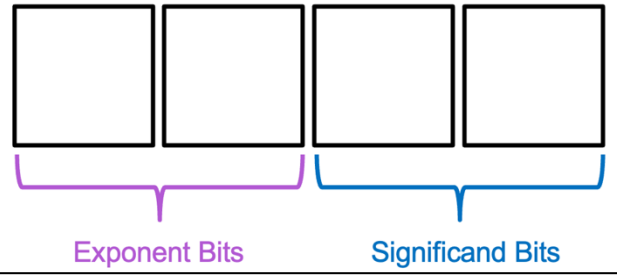
3

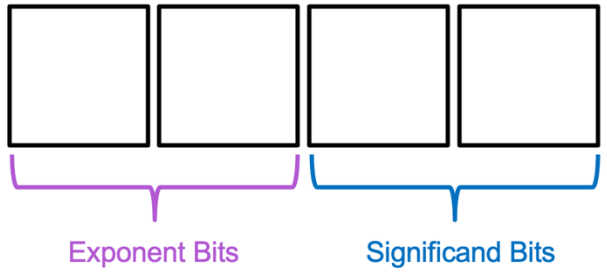
Microfloat V4

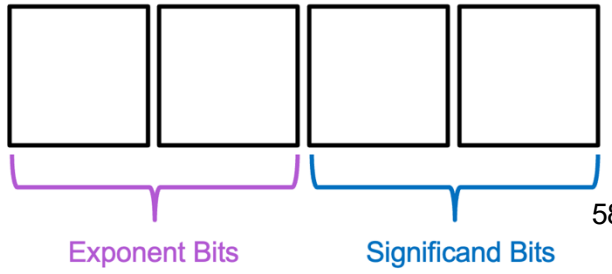
- For each value below, determine if it can be represented.
  - If it can be represented, write out its Microfloat V4 encoding.
  - If not, explain why: not enough precision or not enough range.
- A) 0.5
- B) 1.75
- C) 3.5
- D) 4
- E) 0.125











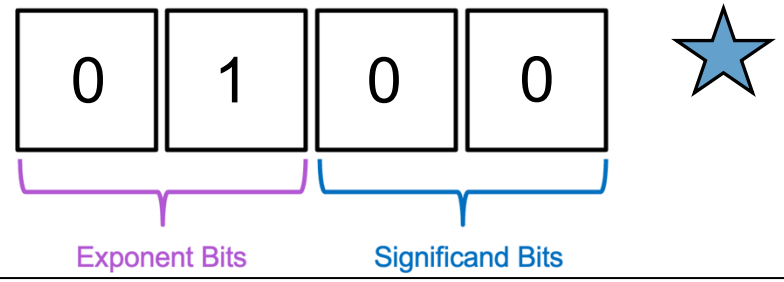
Microfloat V4

- For each value below, determine if it can be represented.
  - If it can be represented, write out its Microfloat V4 encoding.
  - If not, explain why: not enough precision or not enough range.
- A) $0.5 = 1_2 \times 2^{-1} = 0100_2$
- B) $1.75 = 1.11_2 \times 2^0 = 1011_2$
- C) $3.5 = 1.11 \times 2^1 = 1111_2$
- D) $4 = 1_2 \times 2^2 \Rightarrow$ not enough range
- E) $0.125 = 1_2 \times 2^{-3}$

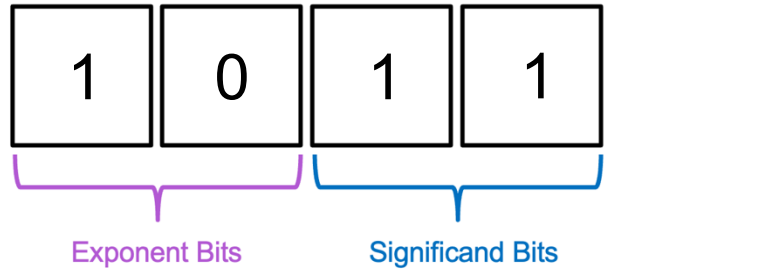
  $\Rightarrow$ not enough range

Exponent is -3 + 2 = -1. Exponent must be positive

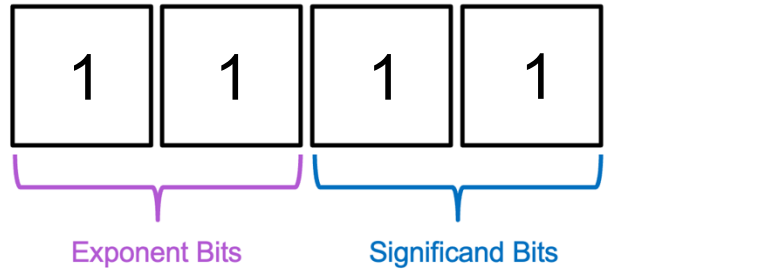
A



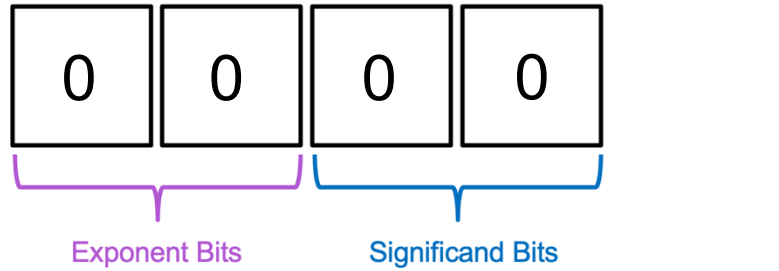
B



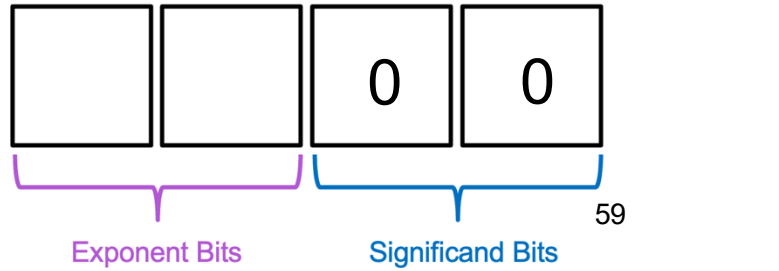
C



D



E



Microfloat V4

Exponent	Significand	Value	Decimal
00_2	00_2	$1.00_2 \times 2^{0-2}$	0.25
00_2	01_2	$1.01_2 \times 2^{0-2}$	0.3125
00_2	10_2	$1.10_2 \times 2^{0-2}$	0.375
00_2	11_2	$1.11_2 \times 2^{0-2}$	0.4375
01_2	00_2	$1.00_2 \times 2^{1-2}$	0.5
01_2	01_2	$1.01_2 \times 2^{1-2}$	0.625
01_2	10_2	$1.10_2 \times 2^{1-2}$	0.75
01_2	11_2	$1.11_2 \times 2^{1-2}$	0.875
10_2	00_2	$1.00_2 \times 2^{2-2}$	1
10_2	01_2	$1.01_2 \times 2^{2-2}$	1.25
10_2	10_2	$1.10_2 \times 2^{2-2}$	1.5
10_2	11_2	$1.11_2 \times 2^{2-2}$	1.75
11_2	00_2	$1.00_2 \times 2^{3-2}$	2
11_2	01_2	$1.01_2 \times 2^{3-2}$	2.5
11_2	10_2	$1.10_2 \times 2^{3-2}$	3
11_2	11_2	$1.11_2 \times 2^{3-2}$	3.5

Distance between consecutive values

0.0625

0.125

0.25

0.5